

10.0 FLOW CONTROL

The ENC28J60 implements hardware flow control for both Full and Half-Duplex modes. The operation of this feature differs depending on which mode is being used.

10.1 Half-Duplex Mode

In Half-Duplex mode, setting the EFLOCON.FCEN0 bit causes flow control to be enabled. When FCEN0 is set, a continuous preamble pattern of alternating '1's and '0's (55h) will automatically be transmitted on the Ethernet medium. Any connected nodes will see the transmission and either not transmit anything, waiting for the ENC28J60's transmission to end, or will attempt to transmit and immediately cause a collision. Because a collision will always occur, no nodes on the network will be able to communicate with each other and no new packets will arrive.

When the host controller tells the ENC28J60 to transmit a packet by setting ECON1.TXRTS, the preamble pattern will stop being transmitted. An Inter-Packet Gap delay will pass as configured by register MABBIPG and then the ENC28J60 will attempt to transmit its packet. During the Inter-Packet Gap delay, other nodes may begin to transmit. Because all traffic was jammed previously, several nodes may begin transmitting and a series of collisions may occur. When the ENC28J60 successfully finishes transmitting its packet or aborts it, the transmission of the preamble pattern will automatically restart. When the host controller wishes to no longer jam the network, it should clear the FCEN0 bit. The preamble transmission will cease and normal network operation will resume.

Given the detrimental network effects that are possible and lack of effectiveness, it is not recommended that half-duplex flow control be used unless the application will be in a closed network environment with proper testing.

10.2 Full-Duplex Mode

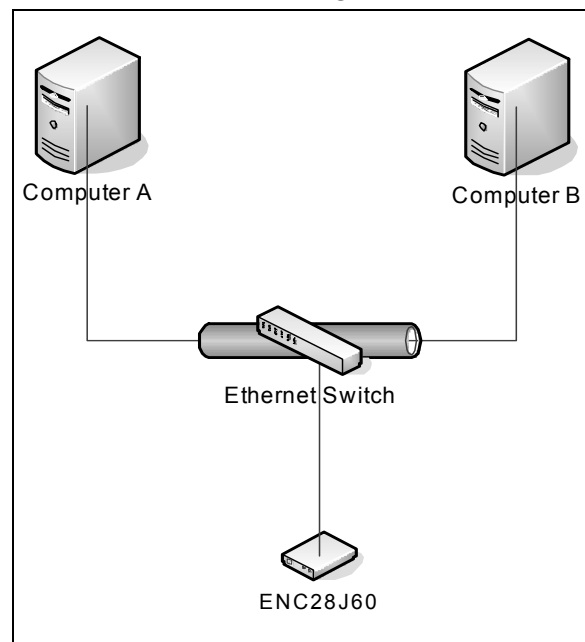
In Full-Duplex mode (MACON3.FULDPX = 1), hardware flow control is implemented by means of transmitting pause control frames as defined by the IEEE 802.3 specification. Pause control frames are 64-byte frames consisting of the reserved Multicast destination address of 01-80-C2-00-00-01, the source address of the sender, a special pause opcode, a 2-byte pause timer value and padding/CRC.

Normally, when a pause control frame is received by a MAC, the MAC will finish the packet it is transmitting and then stop transmitting any new frames. The pause timer value will be extracted from the control frame and used to initialize an internal timer. The timer will auto-

matically decrement every 512 bit times or 51.2 μ s. While the timer is counting down, reception of packets is still enabled. If new pause frames arrive, the timer will be reinitialized with the new pause timer value. When the timer reaches zero or was sent a frame with a zero pause timer value, the MAC that received the pause frame will resume transmitting any pending packets. To prevent a pause frame from stopping all traffic on the entire network, Ethernet switches and routers do not propagate pause control frames in Full-Duplex mode. The pause operation only applies to the recipient.

A sample network is shown in Figure 10-1. If Computer A were to be transmitting too much data to the ENC28J60 in Full-Duplex mode, the ENC28J60 could transmit a pause control frame to stop the data which is being sent to it. The Ethernet switch would take the pause frame and stop sending data to the ENC28J60. If Computer A continues to send data, the Ethernet switch will buffer the data so it can be transmitted later when its pause timer expires. If the Ethernet switch begins to run out of buffer space, it will likely transmit a pause control frame of its own to Computer A. If, for some reason, the Ethernet switch does not generate a pause control frame of its own, or one of the nodes does not properly handle the pause frame it receives, then packets will inevitably be dropped. In any event, any communication between Computer A and Computer B will always be completely unaffected.

FIGURE 10-1: SAMPLE FULL-DUPLEX NETWORK



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To enable flow control on the ENC28J60 in Full-Duplex mode, the host controller must set the TXPAUS and RXPAUS bits in the MACON1 register. Then, at any time that the receiver buffer is running out of space, the host controller should turn flow control on by writing the value 02h to the EFLOCON register. The hardware will periodically transmit pause frames loaded with the pause timer value specified in the EPAUS registers. The host controller can continue to transmit its own packets without interfering with the flow control hardware.

When space has been made available for more packets in the receive buffer, the host controller should turn flow control off by writing the value 03h to the EFLOCON register. The hardware will send one last pause frame loaded with a pause timer value of 0000h. When the pause frame is received by the remote node, it will resume normal network operations.

When RXPAUS is set in the MACON1 register and a valid pause frame arrives with a non-zero pause timer value, the ENC28J60 will automatically inhibit transmissions. If the host controller sets the ECON1.TXRTS bit to send a packet, the hardware will simply wait until the pause timer expires before attempting to send the packet and subsequently clearing the TXRTS bit. Normally, the host controller will never know that a pause frame has been received. However, if it is desirable to the host controller to know when the MAC is paused or not, it should set the PASSALL bit in MACON1 and then manually interpret the pause control frames which may arrive.

REGISTER 10-1: EFLOCON: ETHERNET FLOW CONTROL REGISTER

U-0	U-0	U-0	U-0	U-0	R-0	R/W-0	R/W-0
—	—	—	—	—	FULDPXS	FCEN1	FCEN0
bit 7					bit 0		

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-3 **Unimplemented:** Read as '0'

bit 2 **FULDPXS:** Read-Only MAC Full-Duplex Shadow bit

1 = MAC is configured for Full-Duplex mode, FULDPX (MACON3<0>) is set

0 = MAC is configured for Half-Duplex mode, FULDPX (MACON3<0>) is clear

bit 1-0 **FCEN1:FCEN0:** Flow Control Enable bits

When FULDPXS = 1:

11 = Send one pause frame with a '0' timer value and then turn flow control off

10 = Send pause frames periodically

01 = Send one pause frame then turn flow control off

00 = Flow control off

When FULDPXS = 0:

11 = Flow control on

10 = Flow control off

01 = Flow control on

00 = Flow control off

TABLE 10-1: SUMMARY OF REGISTERS USED WITH FLOW CONTROL

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
ECON1	TXRST	RXRST	DMAST	CSUMEN	TXRTS	RXEN	BSEL1	BSEL0	13
MACON1	—	—	—	r	TXPAUS	RXPAUS	PASSALL	MARXEN	14
MABBIPG	—	Back-to-Back Inter-Packet Gap (BBIPG<6:0>)							14
EFLOCON	—	—	—	—	—	FULDPXS	FCEN1	FCEN0	14
EPAUSL	Pause Timer Value Low Byte (EPAUS<7:0>)								14
EPAUSH	Pause Timer Value High Byte (EPAUS<15:8>)								14

Legend: — = unimplemented, read as '0'. Shaded cells are not used.

ENC28J60

NOTES: